Joint Estimation of Duration-Implied Risk-Free Rate and Bond Risk Premium

A Multidisciplinary Treatise

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Abstract

A continuous-time coupon bond has a price $P(t)$ and a parallel-shift (Fisher–Weil) duration $D(t)$. Suppose the observed quantity is $Q(D) = P(D)(1+D)/(1+r_f(D)+p(D))$. The function $p(D)$ is then an *implied risk premium*, defined by inverting the pricing relation, in the same sense that an implied volatility is defined by inverting an option formula. Only the sum $r_f(D)+p(D)$ is identified by a cross-section of prices. This treatise (i) derives the price and duration of the bond from first principles, (ii) proves that the implied sum is a closed-form functional of the cash flows and of Q, (iii) proves the observational equivalence of splits of the sum and states sharp conditions under which a joint split is identified when a risk-free sample and a risky sample share a curve, (iv) constructs a joint nonlinear least-squares estimator with a Nelson–Siegel risk-free curve and a basis expansion for the premium, reduces it by variable projection, and derives its Jacobian, asymptotic distribution and inference, (v) gives structural foundations for $p(D)$ in affine term-structure, reduced-form credit, liquidity and stochastic-discount-factor models, and (vi) interprets the construction through mathematics, logic, economics, financial planning, econometrics, law, the natural sciences, the behavioural sciences, philosophy, religion, political science, international relations, welfare and warfare economics, statistics, computer science, data science, machine learning and cybersecurity. A bibliography, a glossary and a reference implementation are included.

The treatise ends with "The End"

Contents

1 Introduction **3**

2 Notation and conventions **4**

3 The continuous-time bond: price and duration **5**
3.1 Cash-flow measure and discounting 5
3.2 Duration 5
3.3 Constant rate and constant coupon 7
3.4 Comparison with other duration concepts 8

4 The implied premium **8**
4.1 Definition and inversion 8
4.2 Cash-flow representation 9
4.3 An alternative reading 9

5 Identification **10**

6 The joint estimator **11**
6.1 The risk-free curve . . . 11
6.2 Model quantities as functions of θ . . . 11
6.3 The risk premium basis . . . 11
6.4 Error structure, and two criteria . . . 12
6.5 Analytic Jacobian . . . 12
6.6 Variable projection . . . 13
6.7 Algorithm . . . 13
6.8 Complexity . . . 15

7 Asymptotic theory and inference **15**
7.1 Inference on $\hat{p}(D)$ . . . 15

8 Structural foundations of the premium **16**
8.1 A stochastic-discount-factor bound . . . 16
8.2 Gaussian affine term structure . . . 16
8.3 Reduced-form credit risk . . . 17
8.4 Liquidity and demand . . . 17

9 Simulation and empirical protocol **17**

10 Interdisciplinary perspectives **18**
10.1 Mathematics and logic . . . 18
10.2 Economics and the theory of finance . . . 19
10.3 Financial planning and analysis . . . 19
10.4 Econometrics . . . 19
10.5 Law . . . 19
10.6 Physics . . . 20
10.7 Chemistry . . . 20
10.8 Biology and pharmacology . . . 20
10.9 Linguistics . . . 20
10.10 Psychology . . . 20
10.11 Psychiatry and neuroscience . . . 21
10.12 Philosophy . . . 21
10.13 Religion . . . 21
10.14 Political science . . . 21
10.15 Geopolitics . . . 21
10.16 Diplomacy . . . 21
10.17 International relations . . . 22
10.18 Welfare economics . . . 22
10.19 Warfare economics . . . 22
10.20 Warfare . . . 22
10.21 Statistics . . . 22
10.22 Computer science and algorithms . . . 23
10.23 Data science . . . 23
10.24 Artificial intelligence and machine learning . . . 23
10.25 Cybersecurity . . . 23
10.26 Further fields . . . 23

11 Limitations and research agenda **24**

12 Conclusion **24**

A Additional derivations **24**
A.1 Duration of a continuous coupon with exponential growth 24
A.2 Convexity of the constant case 25
A.3 Why the equation-error regression is inconsistent 25
A.4 Sensitivity of the premium to the curve 25

B Reference implementation (Python) **25**

List of Figures

1 Cash-flow measure on $[t, T]$. Duration is the mean of $s - t$ under present-value weights. 7
2 Duration as a centre of mass. Annual-coupon discretisation of the example in Table 2 with $\tau = 5$, $r = 4\%$, $C = 0.05$ paid at $k = 1, \ldots, 5$ (final bar includes principal). The weights are w_k in %; heights are shown at 0.06 cm per percentage point. The balance point of the bars is $D \approx 4.56$ years. 8
3 Flow of the variable-projection estimator. 14
4 Stylised decomposition. Circles: sovereign sample S (on r_f). Squares: risky sample K (on $g = r_f + p$). The vertical gap on the overlap of supports identifies $p(D)$ (Proposition 5.5). Curves are illustrative and not estimated from data. . . . 14

List of Tables

1 Principal notation. 4
2 Constant $r = 4\%$, coupon $C = 0.05$, $M = 1$: price, duration and scaled price (computed from (12)–(13)). The perpetual limit of D is $1/r = 25$. 7
3 Duration concepts and the one used here. 8
4 Identifying restrictions and what they buy. 11
5 Monte Carlo design (a specification, not a reported result). 17
6 Empirical pipeline and diagnostics. 18
7 Map of fields to the roles of duration D, premium p and joint estimation. 18
8 Uses of $\hat{r}_f(D)$ and $\hat{p}(D)$ in planning and analysis. 19

1 Introduction

Fixed-income valuation separates a *discount rate* into a component that compensates for the time value of money and a component that compensates for bearing risk. The first component is conventionally proxied by a government curve and the second is called a spread or a premium [58, 30, 42]. In practice only the total discount rate is visible, because only prices are observed. Any decomposition therefore rests on a model or on an identifying restriction, and it inherits all the weaknesses of the joint hypothesis problem [52].

This treatise starts from a specific, compact pricing relation. A bond with cash flows $C(\cdot)$ and terminal payment M has a model price P and a duration D. The observed quantity is

$$Q(D) = \frac{P(D)\,(1 + D)}{1 + r_f(D) + p(D)}, \tag{1}$$

and $p(D)$ is the *implied risk premium*. The duration concept used throughout is the continuous-time parallel-shift duration of Fisher and Weil [59], which is neither the Macaulay duration [112] nor the modified duration of practitioner texts [51].

Contributions.

1. A self-contained derivation of $P(t)$ and $D(t)$ with the law of motion $\dot{D} = -1 + DC/P$, with proofs (Section 3).
2. A closed-form cash-flow expression for $r_f + p$ and an explicit statement of the scale convention hidden in (1) (Section 4).
3. A sharp identification analysis for the joint problem (Section 5).
4. An estimator, its variable-projection reduction, analytic Jacobian, asymptotic theory and inference (Sections 6–7).
5. Structural microfoundations for $p(D)$ that suggest which basis functions to use (Section 8).
6. A design for simulation and empirical work (Section 9).
7. Interpretations in more than thirty fields (Section 10).

Roadmap. Section 2 fixes notation. Sections 3–5 are theory. Sections 6–7 are estimation. Section 8 gives microfoundations. Section 9 is the empirical protocol. Section 10 is interdisciplinary. Section 11 lists limitations. The appendices contain additional proofs and code.

2 Notation and conventions

Table 1: Principal notation.

Symbol	Meaning
t, T, $\tau = T - t$	valuation date, maturity date, time to maturity (years)
$r(u)$	instantaneous risk-free short rate
$Z(t, s)$	discount factor $\exp\big(-\int_t^s r(u)\,\mathrm{d}u\big)$
$C(s)$	continuous coupon rate (currency per year)
M	terminal (face) payment at T
μ	cash-flow measure $C(s)\mathrm{d}s + M\delta_T$
$P(t)$	model price (risk-free discounting)
$D(t)$	parallel-shift (Fisher–Weil) duration
$\widetilde{P}$	duration-scaled price $P(1 + D)$
Q	observed (duration-scaled) market quantity in (1)
$r_f(D)$	risk-free rate at duration D
$p(D)$	implied risk premium at duration D
$g = r_f + p$	implied total discount rate ("implied sum")
$y(m; \theta)$	zero-coupon yield curve at maturity m with parameters θ
S, K	sovereign (risk-free) sample, risky sample
φ	coefficients of the premium basis $b_k(\cdot)$
$\psi = (\theta, \varphi)$	stacked parameter vector

Remark 2.1 (Units and the scale of Q). In (1) the quantity D is measured in years and added to the number one, and r_f, p are annualised rates. The expression is therefore a *normalisation convention*, not a dimensionally homogeneous law. Its consequence is that Q must be understood on the same duration-scaled footing as $P(1 + D)$. If a raw market price $\Pi \approx P$ were inserted

for Q, the identity $g = P(1+D)/\Pi - 1$ would return $g \approx D$, i.e. a "rate" of several hundred percent for a ten-year bond. We therefore carry an indicator $\iota \in \{0,1\}$ and write

$$\widetilde{P}^{(\iota)} = P\,(1+\iota D), \tag{2}$$

so that $\iota = 1$ is the user-specified convention and $\iota = 0$ is the plain relation $Q = P/(1+r_f+p)$ for raw prices. Everything below holds for both values.

3 The continuous-time bond: price and duration

3.1 Cash-flow measure and discounting

Fix $t \in [0,T]$. Assume $r : [0,T] \to \mathbb{R}$ and $C : [0,T] \to [0,\infty)$ are continuous. The bond pays coupons at rate $C(s)$ on $[t,T]$ and the amount $M > 0$ at T. Define the discount factor

$$Z(t,s) = \exp\Big(-\int_t^s r(u)\,\mathrm{d}u\Big), \qquad t \le s \le T. \tag{3}$$

The cash flows form the measure $\mu(\mathrm{d}s) = C(s)\,\mathrm{d}s + M\,\delta_T(\mathrm{d}s)$ on $[t,T]$.

Definition 3.1 (Price)**.** The model price is

$$P(t) = \int_{[t,T]} Z(t,s)\,\mu(\mathrm{d}s) = \int_t^T C(s)Z(t,s)\,\mathrm{d}s + M\,Z(t,T). \tag{4}$$

Proposition 3.2 (Price dynamics)**.** *The price P is continuously differentiable and solves*

$$\frac{\mathrm{d}P}{\mathrm{d}t} = r(t)P(t) - C(t), \qquad P(T) = M. \tag{5}$$

Proof. From (3), $\partial_t Z(t,s) = r(t)Z(t,s)$ and $Z(t,t) = 1$. Differentiating (4) with Leibniz' rule,

$$\begin{aligned}\frac{\mathrm{d}P}{\mathrm{d}t} &= -C(t)Z(t,t) + \int_t^T C(s)\,\partial_t Z(t,s)\,\mathrm{d}s + M\,\partial_t Z(t,T)\\ &= -C(t) + r(t)\Big[\int_t^T C(s)Z(t,s)\,\mathrm{d}s + MZ(t,T)\Big] = r(t)P(t) - C(t).\end{aligned}$$

At $t = T$ the integral vanishes and $Z(T,T) = 1$, so $P(T) = M$. Uniqueness follows because (5) is linear with continuous coefficients (Picard–Lindelöf). □

Corollary 3.3 (No-arbitrage interpretation)**.** *Rewriting (5) as $(\mathrm{d}P + C\,\mathrm{d}t)/P = r\,\mathrm{d}t$ shows that the instantaneous total return (capital gain plus coupon yield) of the bond equals the risk-free rate.*

Remark 3.4 (Stochastic rates). If r is an Itô process then $P(t) = \mathbb{E}_t^{\mathbb{Q}}[\int_t^T C(s)Z(t,s)\,\mathrm{d}s + MZ(t,T)]$ and, by the Feynman–Kac theorem [144], $P(t,r)$ solves $\partial_t P + \mu_r \partial_r P + \tfrac{1}{2}\sigma_r^2 \partial_{rr} P - rP + C = 0$ with $P(T,r) = M$ [154, 35, 22, 57].

3.2 Duration

Introduce a parallel shift ε of the entire discount curve, $r(u) \mapsto r(u) + \varepsilon$. Each discount factor becomes $Z(t,s)e^{-\varepsilon(s-t)}$, and

$$P_\varepsilon(t) = \int_t^T C(s)Z(t,s)e^{-\varepsilon(s-t)}\,\mathrm{d}s + MZ(t,T)e^{-\varepsilon(T-t)}. \tag{6}$$

Definition 3.5 (Parallel-shift duration)**.**

$$D(t) = -\frac{1}{P(t)} \frac{\partial P_\varepsilon(t)}{\partial \varepsilon}\Big|_{\varepsilon=0} = -\frac{\partial \ln P_\varepsilon(t)}{\partial \varepsilon}\Big|_{\varepsilon=0}. \tag{7}$$

Theorem 3.6 (Properties of duration)**.** *Under the standing assumptions, with* $w(\mathrm{d}s) = Z(t,s)\mu(\mathrm{d}s)/P(t)$ *the present-value weights (a probability measure on* $[t,T]$*):*

(i) (Time-weighted average)

$$D(t) = \int (s-t)\, w(\mathrm{d}s) = \frac{1}{P(t)}\Big[\int_t^T (s-t)C(s)Z(t,s)\,\mathrm{d}s + (T-t)MZ(t,T)\Big]. \tag{8}$$

(ii) (Bounds) $0 \le D(t) \le T-t$*, with* $D(t) = T-t$ *if and only if* $C \equiv 0$ *on* $[t,T]$*.*

(iii) (Convexity as variance) $\dfrac{1}{P}\dfrac{\partial^2 P_\varepsilon}{\partial \varepsilon^2}\Big|_0 = \int (s-t)^2 w(\mathrm{d}s)$ *and* $\dfrac{\partial^2 \ln P_\varepsilon}{\partial \varepsilon^2}\Big|_0 = \mathrm{Var}_w(s-t) \ge 0$*. Hence* $\varepsilon \mapsto \ln P_\varepsilon$ *is convex.*

(iv) (Law of motion) D *is absolutely continuous and*

$$\frac{\mathrm{d}D}{\mathrm{d}t} = -1 + D(t)\frac{C(t)}{P(t)}. \tag{9}$$

Proof. (i) Differentiating (6) under the integral sign (dominated convergence; the integrand is smooth in ε and bounded on compacts) gives $\partial_\varepsilon P_\varepsilon|_0 = -\int (s-t) Z\, \mu(\mathrm{d}s)$; divide by $-P$.

(ii) The weights w are positive and sum to one. Their support lies in $[t,T]$, so the mean of $s-t$ lies in $[0, T-t]$. The upper bound is attained exactly when w is the point mass at T, i.e. when no coupon carries positive weight, i.e. $C \equiv 0$.

(iii) Differentiate twice: $\partial_\varepsilon^2 P_\varepsilon = \int (s-t)^2 Z_\varepsilon\, \mu(\mathrm{d}s)$. For the logarithm, $\partial^2 \ln P_\varepsilon = P_{\varepsilon\varepsilon}/P - (P_\varepsilon/P)^2 = \mathbb{E}_w[(s-t)^2] - (\mathbb{E}_w[s-t])^2 = \mathrm{Var}_w(s-t)$ at $\varepsilon = 0$. The same computation at any ε uses the tilted weights, which are again probability weights, so the second derivative is nonnegative everywhere.

(iv) Let $N(t) = \int_t^T (s-t)C(s)Z(t,s)\,\mathrm{d}s + (T-t)MZ(t,T)$, so $D = N/P$. The boundary term at $s = t$ vanishes because $(s-t) = 0$. Differentiating,

$$N'(t) = -\Big[\int_t^T C(s) Z\,\mathrm{d}s + MZ(t,T)\Big] + r(t)N(t) = -P(t) + r(t)N(t),$$

where the first term is from differentiating $(s-t)$ and $(T-t)$, and the second from $\partial_t Z = rZ$. Then

$$D' = \frac{N'P - NP'}{P^2} = \frac{(rN-P)P - N(rP-C)}{P^2} = -1 + \frac{NC}{P^2} = -1 + D\frac{C}{P}.$$

□

Remark 3.7. Equation (9) says that duration shortens at one year per year, except that coupon payment, whose current yield is C/P, pushes duration up by $D\,C/P$ because the bond keeps its more distant cash flows while shedding a near one. For a zero-coupon bond $D' = -1$ and $D = T-t$.

Definition 3.8 (Dollar duration)**.** $P(t)D(t) = -\partial_\varepsilon P_\varepsilon|_0$, the currency loss per unit parallel increase of the curve. Its time derivative is $N' = rN - P$.

3.3 Constant rate and constant coupon

Let r and C be constant, $\tau = T - t$. The two basic integrals are

$$\int_0^\tau e^{-ru}\,\mathrm{d}u = \frac{1 - e^{-r\tau}}{r}, \tag{10}$$

$$\int_0^\tau u\,e^{-ru}\,\mathrm{d}u = \Big[-\frac{u}{r}e^{-ru}\Big]_0^\tau + \frac{1}{r}\int_0^\tau e^{-ru}\,\mathrm{d}u = \frac{1 - e^{-r\tau}(1 + r\tau)}{r^2}, \tag{11}$$

the second by integration by parts. Hence

$$P = \frac{C}{r}\big(1 - e^{-r\tau}\big) + Me^{-r\tau}, \tag{12}$$

$$D = \frac{1}{P}\Big[\frac{C\big(1 - e^{-r\tau}(1 + r\tau)\big)}{r^2} + \tau M e^{-r\tau}\Big]. \tag{13}$$

Corollary 3.9. *(a) Zero coupon ($C = 0$): $D = \tau$. (b) Perpetual limit with $C > 0$: $D \to 1/r$ as $\tau \to \infty$. (c) Par bond ($C = rM$): $P = M$ and $D = (1 - e^{-r\tau})/r$.*

Proof. (a) is immediate. (b) As $\tau \to \infty$, $e^{-r\tau} \to 0$ and $\tau e^{-r\tau} \to 0$, so $P \to C/r$ and the bracket tends to C/r^2; the ratio is $1/r$. (c) With $C = rM$, $P = M(1 - e^{-r\tau}) + Me^{-r\tau} = M$. The bracket equals $M[(1 - e^{-r\tau}(1 + r\tau))/r + \tau e^{-r\tau}] = M(1 - e^{-r\tau})/r$; divide by $P = M$. □

Table 2: Constant $r = 4\%$, coupon $C = 0.05$, $M = 1$: price, duration and scaled price (computed from (12)–(13)). The perpetual limit of D is $1/r = 25$.

τ (years)	P	D	$P(1 + D)$
1	1.0098	0.9756	1.9949
5	1.0453	4.4400	5.6866
10	1.0824	7.9698	9.7091
30	1.1747	16.6671	20.7534

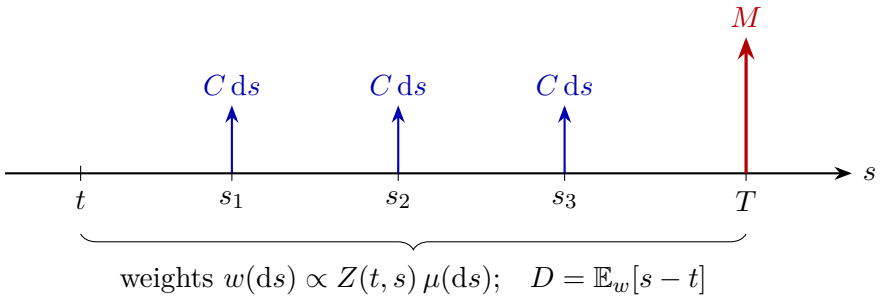


Figure 1: Cash-flow measure on $[t, T]$. Duration is the mean of $s - t$ under present-value weights.

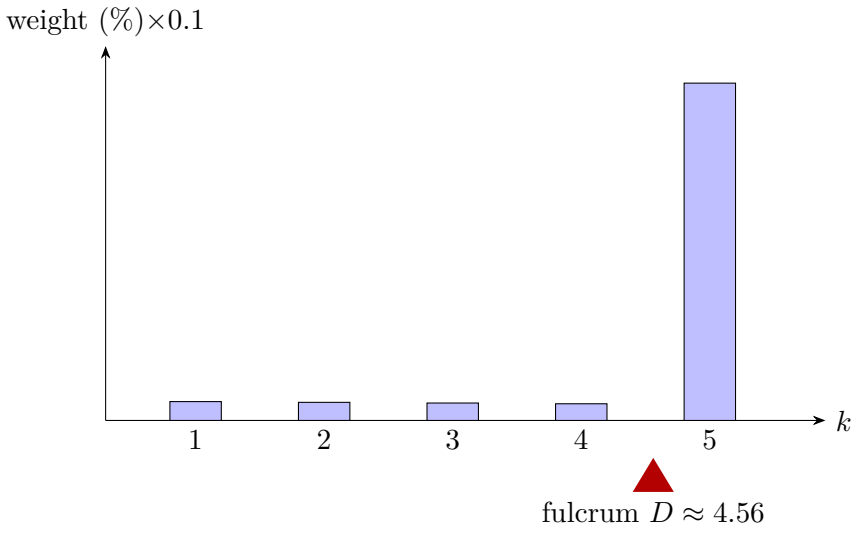


Figure 2: Duration as a centre of mass. Annual-coupon discretisation of the example in Table 2 with $\tau = 5$, $r = 4\%$, $C = 0.05$ paid at $k = 1, \dots, 5$ (final bar includes principal). The weights are w_k in %; heights are shown at 0.06 cm per percentage point. The balance point of the bars is $D \approx 4.56$ years.

3.4 Comparison with other duration concepts

Table 3: Duration concepts and the one used here.

Concept	Definition	Remark
Macaulay [112]	PV-weighted mean time at a single yield y	flat-yield; discrete or continuous compounding
Modified	$D_{\mathrm{Mac}}/(1 + y/m)$	price sensitivity to y with m compounding periods
Effective	$-(P_+ - P_-)/(2P\,\Delta y)$	numerical; handles optionality
Fisher–Weil [59]	PV-weighted mean time at curve discounting	parallel shift of the *curve*; equals (8)
Key-rate [84]	sensitivity to local curve bumps	decomposes D along maturities
Factor [108]	loadings on principal components	level, slope, curvature

Under continuous compounding the parallel-shift duration and the time-weighted average coincide, and the distinction between the Macaulay and modified duration (a factor $1/(1 + y/m)$) vanishes as $m \to \infty$ [82, 132].

4 The implied premium

4.1 Definition and inversion

Definition 4.1 (Implied total discount rate and implied premium)**.** For a bond with observed $Q > 0$ and model quantities $\widetilde{P} = P(1 + \iota D) > 0$, the *implied total discount rate* is

$$g = r_f + p = \frac{\widetilde{P}}{Q} - 1, \tag{14}$$

and, given a risk-free rate r_f, the *implied risk premium* is

$$p = \frac{\widetilde{P}}{Q} - 1 - r_f. \tag{15}$$

Proposition 4.2 (Well-posedness of the inversion)**.** *For fixed $\widetilde{P} > 0$ and r_f, the map $p \mapsto \hat{Q}(p) = \widetilde{P}/(1 + r_f + p)$ is a strictly decreasing diffeomorphism from $(-1 - r_f, \infty)$ onto $(0, \infty)$ with inverse (15). Moreover*

$$\frac{\partial p}{\partial Q} = -\frac{\widetilde{P}}{Q^2} = -\frac{1+g}{Q}, \qquad \frac{\partial \ln Q}{\partial p} = -\frac{1}{1+g}. \tag{16}$$

Proof. $\hat{Q}'(p) = -\widetilde{P}/(1+r_f+p)^2 < 0$ on the domain, and the limits are $+\infty$ at $p \downarrow -1-r_f$ and 0 at $p \to \infty$, so $\hat{Q}$ is a bijection onto $(0, \infty)$. Solving $Q = \widetilde{P}/(1+r_f+p)$ gives (15). Differentiating (15) in Q gives the first formula in (16), and the second follows from $\ln Q = \ln \widetilde{P} - \ln(1+g)$. □

Remark 4.3 (Spread duration). The quantity $1/(1+g) = Q/\widetilde{P}$ is the price elasticity to a premium shift, the analogue of a "spread duration" in this normalisation. A premium widening Δp lowers Q by approximately $\Delta p/(1+g)$ in relative terms. For small g, $\ln(1+g) \approx g$ gives $p \approx \ln(\widetilde{P}/Q) - r_f$.

4.2 Cash-flow representation

Theorem 4.4 (Closed form of the implied sum)**.** *With $\iota = 1$,*

$$r_f(D) + p(D) = \frac{1}{Q}\left[\int_t^T (1+s-t)\, C(s)\, e^{-\int_t^s r(u)\,\mathrm{d}u}\,\mathrm{d}s + (1+T-t)\, M\, e^{-\int_t^T r(u)\,\mathrm{d}u}\right] - 1. \tag{17}$$

For constant r, C and $\tau = T - t$,

$$r_f + p = \frac{1}{Q}\left[\frac{C}{r}\big(1 - e^{-r\tau}\big) + \frac{C\big(1 - e^{-r\tau}(1+r\tau)\big)}{r^2} + (1+\tau) M e^{-r\tau}\right] - 1. \tag{18}$$

Proof. $P(1+D) = P + PD$. By (4) and (8), $P + PD = \int [1 + (s-t)] Z\, \mu(\mathrm{d}s)$, which is the bracket in (17). Substituting in (14) gives the claim. For constant coefficients, use (10)–(11). □

The market quantity Q cannot be eliminated: p is by construction the residual that reconciles P with Q. Equation (17) depends on $t, T, C(\cdot), M$ and the discount curve $r(\cdot)$, and is a statement about the *model* side of the ratio.

Example 4.5 (Numerical inversion)**.** Take the $\tau = 5$ bond in Table 2 and a flat risk-free rate $r_f = 4\%$. Then $\widetilde{P} = 5.6866$. If $Q = 5.4158$, then $g = 5.6866/5.4158 - 1 = 5.00\%$ and $p = 1.00\%$. A rise of Q to $5.6866/1.04 = 5.4679$ would give $p = 0$.

4.3 An alternative reading

If $r_f + p$ is instead read as a flat continuously compounded yield y pricing the bond at Q, then y solves $Q = \int_t^T C e^{-y(s-t)}\,\mathrm{d}s + M e^{-y(T-t)}$. This is a monotone equation in y (the right side is strictly decreasing) with a unique root, but no closed form except for zero coupon bonds, where $y = \ln(M/Q)/(T-t)$. This is the yield-to-maturity relation, and it corresponds to $\iota = 0$ with a flat curve.

5 Identification

Proposition 5.1 (Observational equivalence)**.** *Suppose $\widetilde{P}_i$, D_i are fixed numbers computed from an externally supplied discount curve. For any function h of duration, the pairs (r_f, p) and $(r_f + h,\, p - h)$ generate the same $\hat{Q}_i = \widetilde{P}_i/(1 + r_f(D_i) + p(D_i))$ for all i. Hence only $g = r_f + p$ is identified from a cross-section of prices.*

Proof. The denominator $1 + (r_f + h) + (p - h) = 1 + r_f + p$ is unchanged and nothing else depends on the split. □

Proposition 5.1 is a direct instance of the general principle that a model with a non-injective parameter-to-distribution map is not identified [137]. Identification must therefore come from restrictions. We use three.

Assumption 5.2 (Structure)**.** (A1) *Common curve.* Risky and risk-free bonds share the same risk-free rate function $r_f(D) = y(D;\theta)$, where $y(m;\theta)$ is the zero-coupon yield at maturity m and the model price discounts at y: $Z(t, t+m) = e^{-m\, y(m;\theta)}$.

(A2) *Risk-free sample.* A sample S of bonds has $p \equiv 0$.

(A3) *Parametric premium.* On the risky sample K, $p(D) = \sum_{k=1}^{K} \varphi_k b_k(D)$ for known basis functions b_k.

Remark 5.3 (Why (A1) links $r_f(D)$ to the curve). The expression $r_f(D)$ is a function of duration, while the discount factor $Z(t, s)$ is a function of horizon. A zero-coupon bond has duration equal to its maturity, so identifying the risk-free rate *at duration* D with the zero yield at maturity D is the natural link. A coupon bond of duration D is then assigned the rate of the zero-coupon bond with the same interest-rate sensitivity. This is a modelling choice that we maintain, not a theorem.

Theorem 5.4 (Local identification)**.** *Let $\psi = (\theta, \varphi)$ and let $J(\psi)$ be the Jacobian of $(g_i - y(D_i;\theta) - p_i)_{i \in S \cup K}$ with respect to ψ, where $g_i = \widetilde{P}_i(\theta)/Q_i - 1$ and $p_i = \mathbf{1}_{i \in K} \sum_k \varphi_k b_k(D_i)$. Under Assumption 5.2, a parameter ψ_0 solving the population equations is locally identified if and only if $J(\psi_0)$ has full column rank $\dim\theta + \dim\varphi$.*

Proof. The population equations are $F(\psi) = 0$ with F smooth. By the implicit-function theorem a solution is locally isolated if and only if $\partial F/\partial\psi = J$ has full column rank; conversely, a nontrivial kernel gives a smooth curve of solutions, so identification fails (see [137]). □

Proposition 5.5 (Nonparametric gap identification)**.** *Suppose that, on an interval I of durations, the risk-free function $r_f = y$ is identified from S (for instance, a dense set of zero-coupon sovereigns), and that $\mathbb{E}[g_i - y(D_i) \mid D_i = D] = p(D)$ on K. Then $p(D) = \mathbb{E}[g_i \mid D_i = D, K] - y(D)$ for every $D \in I \cap \mathrm{supp}(D_K)$. The premium is therefore identified exactly on the overlap of the two supports.*

Proof. Take conditional expectations of $g_i = y(D_i) + p(D_i) + \eta_i$ with $\mathbb{E}[\eta_i \mid D_i] = 0$ and subtract the identified y. □

Remark 5.6 (Collinearity trap). If both y and p are given Nelson–Siegel-type shapes with the *same* decay parameter, then within K alone the loadings $\{1, L_1, L_2\}$ are perfectly collinear between them, except through the weak channel by which θ also enters $\widetilde{P}_i$. Specifically, a parallel shift of the curve by h changes $\ln \widetilde{P}_i$ by $-\mathbb{E}_{\tilde{w}_i}[m]\, h$ with weights $\tilde{w}_i \propto (1+m) Z\, \mu$, while a compensating change $-h$ in the premium leaves the denominator unchanged. The Jacobian is thus nearly rank deficient, with condition number inversely proportional to the dispersion of $\mathbb{E}_{\tilde{w}_i}[m]$ across the sample. This is why the sovereign sample is essential: it breaks the symmetry of Proposition 5.1.

Table 4: Identifying restrictions and what they buy.

Restriction	Identifies	Failure mode
$p \equiv 0$ on S	curve parameters θ	sovereign convenience yield contaminates r_f [101]
Common curve (A1)	transfers $\hat{\theta}$ to K	credit-dependent discounting
Parametric p (A3)	φ given $\hat{\theta}$	basis misspecification
Smoothness penalty	stabilises p	oversmoothing hides humps
Factor structure over time	level of p vs. r_f	unstable loadings
Instruments / CDS [110]	splits default vs. liquidity	weak or invalid instruments [145]

6 The joint estimator

6.1 The risk-free curve

We parametrise the zero-coupon yield by the Nelson–Siegel family [119], with $x = m/\kappa$:

$$y(m;\theta) = \theta_0 + \theta_1 L_1(m) + \theta_2 L_2(m), \quad L_1(m) = \frac{1-e^{-x}}{x}, \quad L_2(m) = L_1(m) - e^{-x}. \tag{19}$$

Proposition 6.1 (Origin of the loadings)**.** *y in (19) is the average of the forward-rate curve $f(u) = \theta_0 + \theta_1 e^{-u/\kappa} + \theta_2 (u/\kappa) e^{-u/\kappa}$ over $[0, m]$.*

Proof. $y(m) = m^{-1}\int_0^m f(u)\,\mathrm{d}u$ (the definition of a zero yield in terms of forwards). Compute $m^{-1}\int_0^m e^{-u/\kappa}\mathrm{d}u = \kappa(1-e^{-x})/m = L_1$, and, by parts, $m^{-1}\int_0^m (u/\kappa)e^{-u/\kappa}\mathrm{d}u = m^{-1}\kappa[1 - e^{-x}(1+x)] = L_1 - e^{-x} = L_2$. □

The Svensson extension [147] adds a second hump term. The dynamic version [38] lets θ evolve over time, and a smoothed alternative is used by central banks [69].

6.2 Model quantities as functions of θ

For bond i with discrete or discretised cash flows c_{ik} at times m_{ik} (a quadrature of the continuous coupon stream),

$$P_i(\theta) = \sum_k c_{ik} Z_{ik}, \quad Z_{ik} = e^{-m_{ik} y(m_{ik};\theta)}, \tag{20}$$

$$N_i(\theta) = \sum_k m_{ik} c_{ik} Z_{ik}, \quad D_i(\theta) = N_i / P_i, \tag{21}$$

$$\widetilde{P}_i(\theta) = P_i + \iota N_i. \tag{22}$$

6.3 The risk premium basis

Let $p(D;\varphi) = \sum_k \varphi_k b_k(D)$. Section 8 shows that economically motivated bases are $b = (D)$ (affine models), $b = (1, L_1(D;\kappa_p))$ or $(1, L_1, L_2)$ (mean-reverting hazards), and $b = (D, D^2)$ as a flexible polynomial through the origin. A smoothness penalty can replace parsimony:

Proposition 6.2 (Spline solution of the penalised problem)**.** *Among all $p \in W^{2,2}[D_{\min}, D_{\max}]$, the minimiser of $\sum_{i\in K} w_i^2 (z_i - p(D_i))^2 + \lambda \int (p'')^2 \mathrm{d}D$ is a natural cubic spline with knots at the distinct D_i [135, 156, 37].*

Proof sketch. Let s be the natural cubic spline interpolating the values $\{\hat{p}(D_i)\}$ of any candidate $\hat{p}$. Then $\int (s'')^2 \le \int (\hat{p}'')^2$, because $\int s''(\hat{p}-s)''\mathrm{d}D = 0$ by two integrations by parts, using s''' piecewise constant, $\hat{p}-s=0$ at the knots, and $s''=0$ at the end points. The data-fit term is unchanged, so the penalised objective cannot increase by replacing $\hat{p}$ with s. □

6.4 Error structure, and two criteria

Let the observation equation be $Q_i = \hat{Q}_i(\psi) + \varepsilon_i$, $\mathbb{E}\varepsilon_i = 0$, $\operatorname{Var}\varepsilon_i = \sigma_i^2$, where

$$\hat{Q}_i(\psi) = \frac{\widetilde{P}_i(\theta)}{1 + y(D_i;\theta) + p_i(\varphi)}, \qquad p_i(\varphi) = \mathbf{1}_{i\in K}\sum_k \varphi_k b_k(D_i). \tag{23}$$

Criterion E2 (price errors).

$$\min_{\psi} \sum_{i\in S\cup K} \frac{(Q_i - \hat{Q}_i(\psi))^2}{\sigma_i^2} + \lambda\,\mathcal{R}(\varphi), \tag{24}$$

with $\mathcal{R}$ an optional roughness penalty.

Criterion E1 (implied-sum errors). Define $\hat{g}_i(\theta) = \widetilde{P}_i(\theta)/Q_i - 1$. Then

$$\begin{aligned}\hat{g}_i(\theta) &= \frac{\widetilde{P}_i}{Q_i} - 1 = \frac{\widetilde{P}_i}{\hat{Q}_i + \varepsilon_i} - 1 = (1+g_i)\Big(1 + \frac{\varepsilon_i}{\hat{Q}_i}\Big)^{-1} - 1 \\ &\approx g_i - (1+g_i)\frac{\varepsilon_i}{\hat{Q}_i}.\end{aligned} \tag{25}$$

Hence $\hat{g}_i = y(D_i;\theta) + p_i(\varphi) + \eta_i$ with $\mathbb{E}\eta_i = O(\sigma^2)$ (Jensen) and $\operatorname{Var}\eta_i \approx (1+g_i)^2\sigma_i^2/Q_i^2$. The criterion

$$\min_{\psi} \sum_{i\in S\cup K} w_i^2\big(\hat{g}_i(\theta) - y(D_i;\theta) - p_i(\varphi)\big)^2, \qquad w_i = \frac{Q_i}{\sigma_i(1+g_i)}, \tag{26}$$

is, to first order, equivalent to E2. Its crucial property is that it is *linear in* φ for fixed θ, and its regressors $b_k(D_i)$ contain no noisy market quantity. A naive regression with regressor $Q_i b_k(D_i)$ would be inconsistent (an errors-in-variables problem), which is why (26) is preferred as the inner problem.

Remark 6.3 (Choice of weights). Possible choices for σ_i: the bid–ask half-spread, a multiple of $P_i D_i$ (so that error scales with price risk), or issue-size-based liquidity proxies. The weights w_i depend on g_i; in practice one iterates (iteratively reweighted least squares) with g_i from the previous iterate.

6.5 Analytic Jacobian

Write ∂_j for $\partial/\partial\theta_j$ and note $\partial_j y(m;\theta) = (1, L_1(m), L_2(m))_j$.

Lemma 6.4 (Derivatives of model quantities)**.** *For each bond i (suppress i),*

$$\partial_j P = -\sum_k m_k\,\partial_j y(m_k)\,c_k Z_k, \tag{27}$$

$$\partial_j N = -\sum_k m_k^2\,\partial_j y(m_k)\,c_k Z_k, \tag{28}$$

$$\partial_j D = \frac{\partial_j N - D\,\partial_j P}{P}, \tag{29}$$

$$\partial_j \widetilde{P} = \partial_j P + \iota\,\partial_j N. \tag{30}$$

Proof. $\partial_j Z_k = -m_k \partial_j y(m_k) Z_k$ by the chain rule applied to $Z_k = e^{-m_k y(m_k;\theta)}$; sum over k. The derivative of $D = N/P$ is the quotient rule. □

Proposition 6.5 (Jacobian of the residuals)**.** *For the E1 residual $\rho_i = w_i[\hat{g}_i - y(D_i;\theta) - p_i]$ (treating w_i as fixed),*

$$\frac{\partial \rho_i}{\partial \theta_j} = w_i \Big[\frac{\partial_j \widetilde{P}_i}{Q_i} - \partial_j y(D_i;\theta) - \big(y'(D_i;\theta) + p_i'(D_i)\big)\partial_j D_i \Big], \tag{31}$$

$$\frac{\partial \rho_i}{\partial \varphi_k} = -w_i \, \mathbf{1}_{i\in K} \, b_k(D_i). \tag{32}$$

Here $y' = \partial_m y$ and $p' = \sum_k \varphi_k b_k'$. For E2 the analogous expression is

$$\frac{\partial \hat{Q}_i}{\partial \theta_j} = \frac{\partial_j \widetilde{P}_i}{1+g_i} - \frac{\widetilde{P}_i}{(1+g_i)^2}\big[\partial_j y(D_i) + (y'+p')(D_i)\partial_j D_i\big], \qquad \frac{\partial \hat{Q}_i}{\partial \varphi_k} = -\frac{\widetilde{P}_i \, b_k(D_i)}{(1+g_i)^2}.$$

Proof. Chain rule: D_i depends on θ, and $y(D_i;\theta)$ depends on θ both explicitly and through D_i. Apply Lemma 6.4. □

6.6 Variable projection

For fixed θ, let $u_i = \hat{g}_i(\theta) - y(D_i;\theta)$. On K, $u_K = B\varphi + \eta$, where $B = [b_k(D_i)]_{i\in K}$. Let $W_K = \operatorname{diag}(w_i^2)_{i\in K}$.

Proposition 6.6 (Variable projection [66])**.** *The minimiser of (26) in φ for fixed θ is $\hat{\varphi}(\theta) = (B'W_K B)^{-1} B' W_K u_K$, and the reduced criterion is*

$$\min_\theta \ \Big\| W_S^{1/2} u_S \Big\|^2 + \Big\| W_K^{1/2} (I - H)\, u_K \Big\|^2, \qquad H = B(B'W_K B)^{-1} B' W_K. \tag{33}$$

The minimiser $\hat{\theta}$ of (33) together with $\hat{\varphi}(\hat{\theta})$ minimises (26) jointly.

Proof. For fixed θ, the problem in φ is weighted linear least squares with the stated normal equations. Substituting the solution leaves the residual $(I - H)u_K$ on K and u_S on S (where $p = 0$). Since $\min_{\theta,\varphi} F = \min_\theta \min_\varphi F$, the reduced problem has the same global minimisers in θ. □

6.7 Algorithm

Algorithm 1 (Joint estimation of $r_f(D)$ and $p(D)$).
Input: cash flows $\{(m_{ik}, c_{ik})\}$, observed Q_i, class labels $i \in S$ or K, basis b, grid Θ_κ of decay values, tolerances.

1. For each $\kappa \in \Theta_\kappa$ and each of several starting points $\theta^{(0)}$:
   (a) Compute $Z_{ik}, P_i, N_i, D_i, \widetilde{P}_i$ by (20)–(22).
   (b) Set $\hat{g}_i = \widetilde{P}_i/Q_i - 1$, $u_i = \hat{g}_i - y(D_i;\theta)$, and w_i from the previous iterate.
   (c) Solve $\hat{\varphi} = \arg\min \|W_K^{1/2}(u_K - B\varphi)\|^2$ by QR (not normal equations).
   (d) Form residuals ρ (zero premium on S) and the Jacobian of Proposition 6.5.
   (e) Update θ by a Levenberg–Marquardt step [107, 114]: $(J'J + \lambda_{LM} \operatorname{diag}(J'J))\delta = -J'\rho$.
   (f) Repeat until $\|\delta\|$ and $\|J'\rho\|$ fall below the tolerances.
2. Choose (κ, θ) with the smallest criterion; polish by minimising the price criterion E2 with $\hat{\psi}$ as the starting value.
3. Compute $\hat{V}$ by (35), pointwise standard errors for $\hat{p}(D)$ by (36), and the test (37).

Output: $\hat{r_f}(D) = y(D;\hat{\theta})$, $\hat{p}(D) = b(D)'\hat{\varphi}$, $\hat{g} = \hat{r_f} + \hat{p}$, and $\hat{V}$.

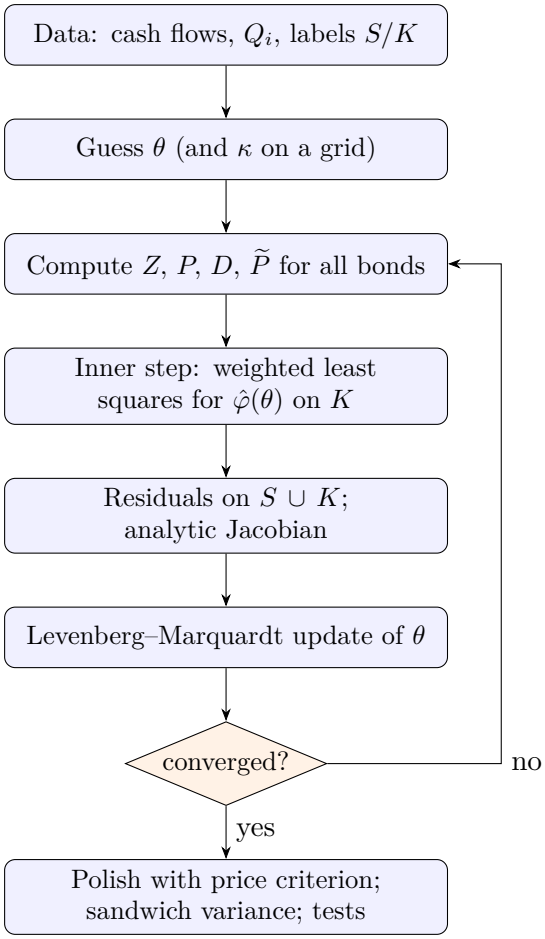


Figure 3: Flow of the variable-projection estimator.

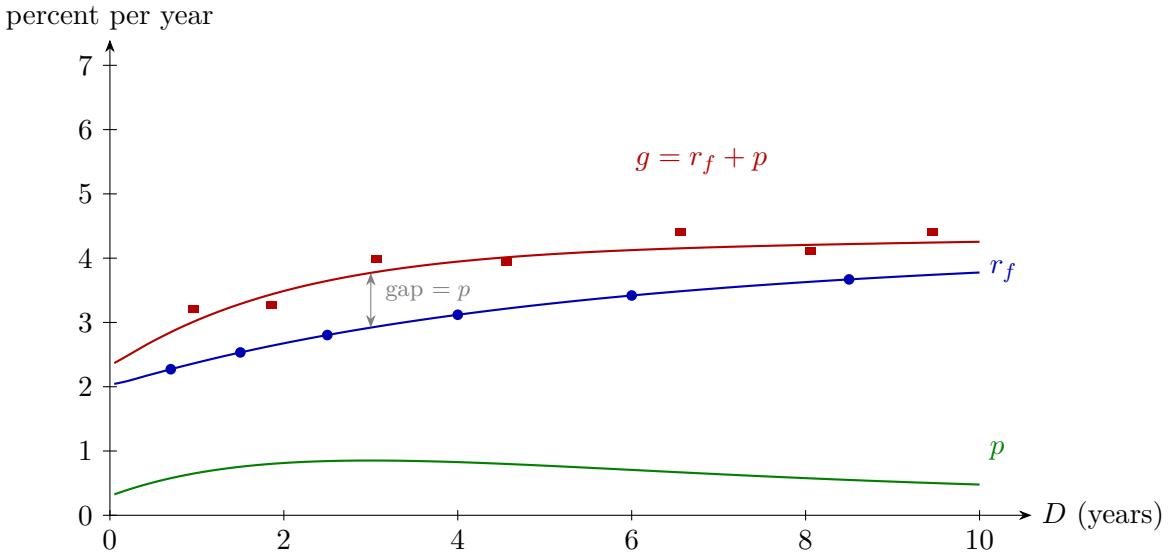


Figure 4: Stylised decomposition. Circles: sovereign sample S (on r_f). Squares: risky sample K (on $g = r_f + p$). The vertical gap on the overlap of supports identifies $p(D)$ (Proposition 5.5). Curves are illustrative and not estimated from data.

6.8 Complexity

With N bonds, n cash flows per bond, $\dim\theta = 3$ and $\dim\varphi = K$, one evaluation of (20)–(22) costs $O(Nn)$ and the analytic Jacobian $O(Nn\dim\theta)$. The inner QR costs $O(NK^2)$ and each LM step a 3×3 solve. The total cost is $O\big(I\,[Nn\cdot 3 + NK^2]\big)$ for I iterations, linear in N [34, 123]. Using a QR factorisation avoids squaring the condition number of B, which matters for polynomial bases.

7 Asymptotic theory and inference

Assumption 7.1 (Regularity)**.** (R1) Ψ is compact and ψ_0 is interior. (R2) Cross-sectional units (bonds or issuer clusters) are independent with finite fourth moments. (R3) The criterion $\mathcal{Q}_N(\psi) = N^{-1}\sum_i \rho_i(\psi)^2$ converges uniformly to a limit uniquely minimised at ψ_0. (R4) $A = \operatorname{plim} N^{-1}J'J$ is positive definite. (R5) ρ_i is twice continuously differentiable.

Theorem 7.2 (Asymptotic normality)**.** *Under Assumption 7.1 [92, 120, 153],*

$$\sqrt{N}(\hat\psi - \psi_0) \xrightarrow{d} \mathcal{N}\big(0,\, A^{-1}BA^{-1}\big), \qquad B = \operatorname{plim} N^{-1}\sum_i \rho_i^2\,\nabla\rho_i\nabla\rho_i'. \tag{34}$$

A consistent estimator is the heteroskedasticity-robust sandwich

$$\hat V = (J'J)^{-1}\Big(\sum_i \hat\rho_i^2\,\hat J_i\hat J_i'\Big)(J'J)^{-1}, \tag{35}$$

where $\hat J_i$ is the ith row of J [158].

Proof. Consistency follows from (R1)–(R3) by the standard extremum-estimator argument. Expand the first-order condition $0 = J(\hat\psi)'\rho(\hat\psi)$ around ψ_0:

$$0 = J(\psi_0)'\rho(\psi_0) + \big[J'J + \textstyle\sum_i \rho_i\nabla^2\rho_i\big](\bar\psi)\,(\hat\psi - \psi_0).$$

At ψ_0, $\mathbb{E}\rho_i \approx 0$, so the second-derivative term is $o_p(N)$. Hence $\sqrt{N}(\hat\psi-\psi_0) = -(N^{-1}J'J)^{-1}N^{-1/2}J'\rho(\psi_0) + o_p(1)$, and the score $N^{-1/2}\sum_i \nabla\rho_i\rho_i$ obeys a Lindeberg–Feller central limit theorem with variance B. Slutsky's theorem yields the result. □

Remark 7.3 (Panels and clustering). With repeated dates, correlation within date and within issuer is typical. Replace the middle term of (35) by cluster sums $\sum_c \hat J_c'\hat\rho_c\hat\rho_c'\hat J_c$ [26], or use Newey–West weighting across dates [121]. The GMM view of the same estimator uses $\mathbb{E}[z_i(\hat g_i - y_i - p_i)] = 0$ with instruments z_i [70].

7.1 Inference on $\hat p(D)$

By the delta method, with $V_{\varphi\varphi}$ the φ-block of $\hat V/N$,

$$\widehat{\mathrm{se}}\big(\hat p(D)\big) = \sqrt{b(D)'\hat V_{\varphi\varphi}\,b(D)}, \qquad \hat p(D) \pm z_{1-\alpha/2}\,\widehat{\mathrm{se}}\big(\hat p(D)\big). \tag{36}$$

For the premium net of the risk-free rate, the joint variance of $\hat g(D) = y(D;\hat\theta) + b(D)'\hat\varphi$ uses the gradient $\nabla_\psi g = (\partial_\theta y(D) + y'(D)\partial_\theta D,\; b(D))$ when D is treated as a function of θ. Simultaneous bands over a grid of durations follow from the sup-t bootstrap [47].

Tests.

1. *No premium:* $H_0 : \varphi = 0$. The Wald statistic
$$\mathcal{W} = \hat{\varphi}' \hat{V}_{\varphi\varphi}^{-1} \hat{\varphi} \xrightarrow{d} \chi^2_K. \tag{37}$$

2. *Sovereign specification:* add a constant offset φ_S to the sovereign sample, $p_S = \varphi_S$. Under $H_0 : \varphi_S = 0$ the estimator is as above. A rejection signals a convenience yield [101] or a misspecified curve. If φ_S is retained, r_f is identified only up to this constant.

3. *Basis adequacy:* compare K via information criteria or time-blocked cross-validation [16, 79].

Remark 7.4 (Bayesian reading). With $\varphi \sim \mathcal{N}(0, \varsigma^2 I)$ and Gaussian noise of variance σ^2, the posterior mean is $\hat{\varphi} = (B'W_K B + \frac{\sigma^2}{\varsigma^2} I)^{-1} B' W_K u_K$, which is ridge regression [85] and a Tikhonov-regularised solution [149]. An ℓ_1 prior gives the lasso [148].

8 Structural foundations of the premium

8.1 A stochastic-discount-factor bound

Let m be a positive stochastic discount factor with $\mathbb{E}[mR] = 1$ for gross returns R [73, 76, 77, 30]. For an excess return R^e, $\mathbb{E}[mR^e] = 0$ implies $\mathbb{E}[R^e] = -\operatorname{Cov}(m, R^e)/\mathbb{E}[m]$. By the Cauchy–Schwarz inequality,
$$\frac{|\mathbb{E}[R^e]|}{\sigma(R^e)} \le \frac{\sigma(m)}{\mathbb{E}[m]}, \tag{38}$$
the Hansen–Jagannathan bound [71]. Since the compensation embedded in an implied premium is a price of bearing exposure to m, $p(D)$ per unit of return volatility is bounded by the volatility of the pricing kernel. Large implied premia relative to volatilities indicate either high risk aversion or a puzzle [115].

8.2 Gaussian affine term structure

Let $\mathrm{d}r = \kappa(\bar{r} - r)\mathrm{d}t + \sigma\,\mathrm{d}W$ under the physical measure $\mathbb{P}$ [154]. Define the market price of risk λ by $\mathrm{d}W = \mathrm{d}W^{\mathbb{Q}} + \lambda\,\mathrm{d}t$. Then under $\mathbb{Q}$, $\mathrm{d}r = [\kappa(\bar{r} - r) + \sigma\lambda]\mathrm{d}t + \sigma\mathrm{d}W^{\mathbb{Q}}$, and the zero-coupon price is $P(t,T) = \exp(A(\tau) - B(\tau)r)$ with $B(\tau) = (1 - e^{-\kappa\tau})/\kappa$.

Proposition 8.1 (Premium linear in rate-duration)**.** *The instantaneous expected excess return of the zero-coupon bond is $\lambda\sigma B(\tau)$. For a coupon bond that is a portfolio of zeros with PV weights w_k, the expected excess return is $\lambda\sigma D_r$, with $D_r = \sum_k w_k B(\tau_k)$ the duration with respect to the short rate.*

Proof. By Itô's lemma and $\mathrm{d}r = (\cdot)\mathrm{d}t + \sigma\mathrm{d}W^{\mathbb{Q}}$, the $\mathbb{Q}$-dynamics of a price satisfying no-arbitrage is $\mathrm{d}P/P = r\,\mathrm{d}t - B\sigma\,\mathrm{d}W^{\mathbb{Q}}$. Substituting $\mathrm{d}W^{\mathbb{Q}} = \mathrm{d}W - \lambda\mathrm{d}t$ gives $\mathrm{d}P/P = (r + \lambda\sigma B)\mathrm{d}t - B\sigma\mathrm{d}W$. The excess return over r is $\lambda\sigma B$. Linear aggregation over zeros with weights w_k gives the coupon-bond result. □

Hence in a one-factor Gaussian model the premium is a straight line through the origin in the (rate-) duration: $p(D) \approx \lambda\sigma D$, which motivates the basis $b = (D)$. With several factors, $p = \sum_j \lambda_j \sigma_j D_j$, and essentially affine specifications allow λ to vary with the state [41, 43, 7]. Empirically, bond risk premia are predicted by forward-rate combinations [31, 53, 28] and can be estimated by regressions [2] or by arbitrage-free models with extra factors [98].

8.3 Reduced-form credit risk

Let default arrive with intensity $h^{\mathbb{Q}}$ under $\mathbb{Q}$ and $h^{\mathbb{P}}$ under $\mathbb{P}$, with fractional loss L of market value at default. Under the recovery-of-market-value convention [44, 90, 91], the risky zero has price $\mathbb{E}^{\mathbb{Q}}[\exp(-\int_t^T (r + h^{\mathbb{Q}} L)\mathrm{d}s)]$. For deterministic r and h, the credit spread over the risk-free zero yield is $\frac{1}{\tau}\int_t^T h^{\mathbb{Q}}(s) L \, \mathrm{d}s$. Decompose

$$p = \underbrace{h^{\mathbb{P}} L}_{\text{expected loss}} + \underbrace{(h^{\mathbb{Q}} - h^{\mathbb{P}})L}_{\text{default risk premium}} + p_{\text{liq}} + p_{\text{tax}} + \dots \tag{39}$$

so that $p_{\text{pure}} = p - h^{\mathbb{P}} L$ is an upper bound on the pure premium unless $p_{\text{liq}}, p_{\text{tax}}$ are modelled [50, 33, 110, 116]. The ratio $h^{\mathbb{Q}}/h^{\mathbb{P}}$ is a jump-risk or clustering factor.

Proposition 8.2 (Mean-reverting hazard gives a Nelson–Siegel premium)**.** *If $h^{\mathbb{Q}}(s) = h_\infty + (h_0 - h_\infty)e^{-s/\kappa_p}$ and L is constant, the average credit spread at horizon m is*

$$p(m) = L\big[h_\infty + (h_0 - h_\infty) L_1(m;\kappa_p)\big], \quad L_1(m;\kappa_p) = \frac{1 - e^{-m/\kappa_p}}{m/\kappa_p}. \tag{40}$$

Proof. $\frac{1}{m}\int_0^m L\, h(s)\mathrm{d}s = L[h_\infty + (h_0 - h_\infty)\kappa_p(1 - e^{-m/\kappa_p})/m]$. This is (40). □

Hence the natural basis for p is $(1, L_1(D;\kappa_p))$, with a possible hump term L_2 if the hazard has a delayed peak. This also explains the collinearity of Remark 5.6: the risk-free curve and the credit premium have the same functional family.

8.4 Liquidity and demand

Bond-specific illiquidity (bid–ask spreads, Amihud-type price impact [5]) and funding constraints [23] add $p_{\text{liq}} = c_0 + c_1 \ell_i$ with ℓ_i an observable liquidity proxy. Preferred-habitat models [155] give a premium that is duration times the product of arbitrageurs' risk aversion and net demand for duration, $p(D) = D\,(a + \gamma' z_t)$, with z_t a demand factor. The convenience yield of safe assets [101] implies $r_f^{\text{sov}} = r_f^* - c(D)$, and $p_S = -c < 0$ in the notation of Remark 2.1.

9 Simulation and empirical protocol

Table 5: Monte Carlo design (a specification, not a reported result).

Element	Choice
True curve	$\theta = (4.5\%, -2.0\%, 1.0\%)$, $\kappa = 2$
True premium	$p(D) = \varphi_1 D + \varphi_2 D^2$, $\varphi = (0.2\%, 0.04\%)$
Sovereigns / corporates	15 / 25 bonds, annual coupons 3% / 5%, maturities 1–15 / 1–12 years
Noise	relative price noise $\sigma = 0.05\%$
Replications	1000, with several random starts
Estimators	E1 with variable projection; E2 polish; naive regression with regressor $Q_i b_k$
Metrics	bias, RMSE of $\hat{p}(D)$ on a grid, coverage of 95% bands, condition number of J
Stress cases	drop S (Remark 5.6); common vs. distinct decay; misspecified b; outliers

Table 6: Empirical pipeline and diagnostics.

Stage	Actions	Diagnostic
Data	prices from trusted venues, cash-flow schedules from indentures	reconciliation of accrued interest
Filters	drop callable, putable, matrix-priced, stale quotes	liquidity screens
Classes	S: sovereign nominal bullet bonds; K: corporates by rating bucket	common-support check in D
Estimation	Algorithm 1 with multiple starts	condition number, rank of J
Inference	clustered sandwich, bootstrap	coverage, Wald test (37)
Validation	rolling-origin out-of-sample pricing error	stability of $\hat{\theta}, \hat{\varphi}$
Governance	version, hash and archive inputs	reproducibility record

10 Interdisciplinary perspectives

Table 7: Map of fields to the roles of duration D, premium p and joint estimation.

Field	Analogue of D	Analogue of p	Main caution
Physics	mean energy $-\partial \ln \mathcal{Z}/\partial\beta$	free-energy offset	analogy only
Chemistry	mean lifetime $1/k$	activation penalty	rate heterogeneity
Biology / pharmacology	mean residence time	hazard of loss	compartment structure
Psychology	subjective delay	dread, probability weighting	duration neglect
Law	contractual maturity	priority, enforceability	jurisdiction
Political science	regime horizon	credible-commitment wedge	endogeneity
Warfare economics	war duration	conflict hazard $\times L$	data scarcity
Welfare economics	social planning horizon	market-versus-social wedge	normative choice
Cybersecurity	data-lifetime exposure	integrity risk premium	adversarial inputs
Machine learning	feature D	learned function $p(\cdot)$	overfitting

10.1 Mathematics and logic

The determination of p from Q is an *inverse problem* in the sense of Hadamard: existence and uniqueness hold pointwise (Proposition 4.2), but the joint problem is ill-posed because uniqueness fails (Proposition 5.1), and stability fails near rank deficiency (Remark 5.6). Regularisation restores well-posedness [149].

Logically, the inference "Q is observed, therefore p is the risk premium" is a conjunction of hypotheses $H = H_P \wedge H_r \wedge H_\iota \wedge H_b$: the cash-flow model, the risk-free curve, the scale convention and the basis are all assumed. If the data refute the fitted model, modus tollens refutes only the conjunction, not any particular conjunct. This is the Duhem–Quine thesis [45, 128] and the finance profession's joint hypothesis problem [52]. Falsifiability [127] is restored only by explicit auxiliary restrictions (such as $p_S = 0$) and out-of-sample tests.

10.2 Economics and the theory of finance

A positive p is the market price of exposure to the discount-factor shocks [30, 42]. Competitive equilibrium and no-arbitrage theory [9, 136, 76, 141] give the structure; the term-structure literature gives the cross-section in D [154, 35, 81]. Time preference itself goes back to Fisher [58] and the discounted-utility form of Samuelson [138]; the information content of prices is the theme of Hayek [80]. Excess volatility [142] and Modigliani–Miller [117] set limits on how much of p can be attributed to cash-flow news and how much to financing structure.

10.3 Financial planning and analysis

An FP&A team uses $r_f + p$ as the market-implied cost of debt in weighted-average cost of capital, hurdle rates, covenant headroom analysis and debt-service budgeting. A premium shock Δp changes the market value of liabilities by $\Delta \ln Q \approx -\Delta p/(1+g)$ (Proposition 4.2), and increases annual interest cost by $B\,\Delta p$ on a balance B at reset.

Asset-liability management. Let assets A and liabilities L have parallel-shift sensitivities. The surplus $\mathcal{S}(\varepsilon) = P_A(\varepsilon) - P_L(\varepsilon)$ satisfies $\mathcal{S}(0) = 0$ when fully funded. Redington immunisation [132] requires

$$\mathcal{S}'(0) = 0 \iff P_A D_A = P_L D_L, \qquad \mathcal{S}''(0) \geq 0 \iff \mathbb{E}_{w_A}[m^2] P_A \geq \mathbb{E}_{w_L}[m^2] P_L, \tag{41}$$

so that every parallel shift weakly increases surplus to second order. This follows by differentiating (6) and using Theorem 3.6(iii). Premium risk (a shift in p) is distinct from rate risk and requires its own hedge or limit, e.g. a credit-spread DV01 equal to $P_i/(1+g_i)$ per unit of p.

Table 8: Uses of $\hat{r_f}(D)$ and $\hat{p}(D)$ in planning and analysis.

Process	Use	Output
Budgeting	forecast cost of debt at refinancing $\hat{r_f} + \hat{p}$	interest budget
Capital allocation	hurdle rate $= \hat{r_f}(D_{\text{proj}}) + \hat{p}(D_{\text{proj}})$	NPV cut-offs
Treasury	duration matching by (41)	hedge ratios
Scenario analysis	shock Δp, $\Delta\theta$	liquidity runway
Valuation	fair-value inputs for illiquid debt [89]	audit-ready model

10.4 Econometrics

Section 7 already gives the asymptotics. Further considerations: (a) D_i is a *generated regressor*, which is why θ and φ are estimated jointly rather than in two steps. (b) Time variation of (θ, φ) is best treated in state-space form: $\psi_t = \psi_{t-1} + \xi_t$, $\hat{g}_{it} = y(D_{it}; \theta_t) + p_{it} + \eta_{it}$, estimated by the Kalman filter [94, 46, 38]. (c) Weak identification (Remark 5.6) yields nonstandard distributions; use identification-robust inference [145]. (d) Outliers are handled by Huber loss [87].

10.5 Law

Cash flows μ are created by contract: the indenture fixes coupon, maturity, covenants and events of default, and in the United States the Trust Indenture Act of 1939 [152] regulates the trustee relationship for qualified public debt. Enforceability and creditor priority determine the loss L in (39). Legal origin and investor protection correlate with credit-market depth and cost [106]. For sovereigns, absence of a bankruptcy court makes political and reputational factors central [151]. Prudential regulation assigns risk weights that become part of p through capital charges [14]. Benchmark manipulation and misreporting are legal wrongs that corrupt the inputs [1].

10.6 Physics

Define $\mathcal{Z}(\beta) = \int e^{-\beta(s-t)} Z(t,s)\mu(\mathrm{d}s)$, i.e. P_ε in (6) with $\beta = \varepsilon$. It is formally a partition function with "energy" $s-t$ and "inverse temperature" ε. Then

$$D = -\frac{\partial \ln \mathcal{Z}}{\partial \beta} = \langle E \rangle, \qquad \frac{\partial^2 \ln \mathcal{Z}}{\partial \beta^2} = \langle E^2 \rangle - \langle E \rangle^2, \tag{42}$$

so duration is the mean energy and convexity-as-variance is the energy fluctuation, which by the fluctuation–dissipation relation is proportional to a heat capacity [133, 25]. The log-convexity of P_ε (Theorem 3.6(iii)) is the thermodynamic stability statement. The pricing PDE is a parabolic equation of diffusion type [18, 19, 113]. The analogy is a source of intuition, not an empirical claim.

10.7 Chemistry

First-order kinetics gives $[A](t) = [A]_0 e^{-kt}$, so $Z(t,s) = e^{-r(s-t)}$ is the survival fraction with rate constant r. Its mean lifetime is $1/r$, the perpetual limit of Corollary 3.9(b). The Arrhenius law $k = Ae^{-E_a/(RT)}$ [8, 10] implies $\partial \ln k / \partial T = E_a/(RT^2)$, a "temperature duration" analogous to sensitivity to a rate shift. A sequence of consecutive reactions (Bateman chains) corresponds to a coupon ladder where each stage feeds the next.

10.8 Biology and pharmacology

In pharmacokinetics the *mean residence time* of a drug is $\mathrm{MRT} = \int_0^\infty t c(t)\mathrm{d}t / \int_0^\infty c(t)\mathrm{d}t$ [159, 63]. This has exactly the form of (8), with concentration playing the role of the present-value cash-flow density. Population models [118] and actuarial survival models [20] show the same structure: a life-contingent annuity is priced as $\int_0^\infty {}_sp_x \, Z(0,s)\,\mathrm{d}s$, where ${}_sp_x$ is a survival function, which is the credit-risky bond formula with the default intensity replaced by a mortality hazard (e.g. Gompertz $h(a) = \alpha e^{\beta a}$).

10.9 Linguistics

"Duration" and "premium" are polysemous terms: Macaulay duration, modified duration, effective duration and the duration of a speech sound share nothing but a word. In the argument of this treatise the terminology is fixed formally (Table 3). Metaphors such as "time is money" structure the concepts [105]. Text analysis provides covariates: finance-specific dictionaries [111] and measures of central-bank communication [72] can be placed in $p(D) = b(D)'\varphi + \gamma' z_t$ as sentiment or uncertainty indices. "Implied" is a technical word: it refers to inversion of a model and not to an assertion that market participants hold a stated belief.

10.10 Psychology

Behavioural research shows non-exponential discounting, e.g. the hyperbolic form $Z_h(m) = (1+\alpha m)^{-\gamma/\alpha}$ [104, 60], whose implied forward rate is not constant and which would be absorbed into the slope of $\hat{r_f}$. Probability weighting in prospect theory [93] overweights small default probabilities, so p exceeds the actuarial $h^{\mathbb{P}}L$ even without risk aversion. Experienced-utility research documents *duration neglect*, where retrospective evaluation ignores the length of an episode [61], a human counterpart to a naive investor who ignores financial duration. Emotion influences risk perception [109].

10.11 Psychiatry and neuroscience

Ventromedial prefrontal damage produces "myopia for the future" in gambling tasks [15], and neuroimaging links anticipatory affect to financial risk-taking: nucleus accumbens activation precedes risk-seeking choices and anterior insula activation precedes risk-averse choices [102]. These findings support treating risk appetite, and hence p, as state-dependent. They do *not* support clinical language about markets: market "mood" is a metaphor, and aggregate prices are not diagnostic instruments for individuals.

10.12 Philosophy

Three issues arise. (i) *Risk versus uncertainty*: Knight [100] and Keynes [97] distinguish measurable risk from unmeasurable uncertainty. An implied premium can contain an *ambiguity premium* [49, 64], not only a risk premium. (ii) *Induction*: estimating p from past prices to predict the future presupposes uniformity of nature [88]. (iii) *Intergenerational ethics*: whether to discount at all, and at which rate, is a moral choice [129, 126, 131]. The Duhem–Quine point of Section 10 on epistemology also applies, as does the instrumentalist defence of unrealistic assumptions [62].

10.13 Religion

Many religious traditions have debated the legitimacy of a positive price of time [86]. In Islamic jurisprudence, *riba* is prohibited, and contemporary Islamic finance uses profit-sharing and asset-backed structures [103]; in such instruments a fixed $r_f + p$ is replaced by a distribution of returns, though conventional curves are often used as pricing benchmarks. For estimation purposes, this implies that $\hat{p}$ cannot be read as "interest" in all legal-religious frameworks, and that a sukuk sample should be treated as a separate class with its own restriction.

10.14 Political science

Institutions that constrain the sovereign's ability to default or expropriate lower the premium. The classic case is England after 1688 [125], and democratic institutions have been argued to give a financial advantage in international competition [140]. A parsimonious specification adds an institutional-quality index INS_i: $p(D) = b(D)'\varphi + \gamma\,\mathrm{INS}_i$. Endogeneity is acute, since the likelihood of default also changes institutions. Political violence and civil conflict raise hazards [32].

10.15 Geopolitics

Geopolitical risk indices [24] supply time-varying covariates for p. Flight-to-safety flows depress sovereign yields; their effect on r_f violates $p_S = 0$ and is handled by the offset φ_S (Section 7). Energy corridors, chokepoints and sanctions regimes affect cash flows (revenue) and not only discount rates, so the cash-flow model P must be revisited and not pushed into p. Historical studies of empire and borrowing costs [56] provide long samples.

10.16 Diplomacy

Diplomatic action is mostly about the beliefs that determine $h^{\mathbb{Q}}$ and L. Treaty guarantees, arbitration clauses and multilateral support reduce loss given default, while a credible commitment to negotiate lowers expected deadweight loss in restructuring. If creditors and a sovereign bargain over a surplus, the Nash bargaining solution gives recovery $R = \alpha V_{\mathrm{res}}$, where V_{res} is the restructured value and $\alpha \in (0, 1)$ the creditor's bargaining power, so that $L = 1 - \alpha V_{\mathrm{res}}/V_0$ [54, 139]. Communication shapes beliefs in ways that can be tracked by text measures [72].

10.17 International relations

Reserve-currency status, alliance structure and the global financial architecture determine which assets form the sovereign sample S [96, 48]. Deviations from covered interest parity, $x = \frac{F}{S}\frac{1+r^*}{1+r} - 1$, are cross-currency wedges of the same functional kind as p [40], and a risk-free curve is therefore currency-specific. Economic statecraft and sanctions [39] are shocks to p of targeted issuers. History warns that sovereign borrowing cycles repeat [134].

10.18 Welfare economics

The social planner's discount rate follows the Ramsey rule $\rho = \delta + \eta g_c$, where δ is pure time preference, η the elasticity of marginal utility and g_c consumption growth [129, 146, 124]. Uncertainty over the rate lowers the effective long-run rate [157]. The wedge p between a market borrowing rate and the social rate creates a deadweight loss. For a small wedge and investment elasticity ϵ_I, a Harberger approximation [75] gives

$$\mathrm{DWL} \approx \tfrac{1}{2}\,\epsilon_I\, I\,\frac{p^2}{1+r_f}. \tag{43}$$

With a public debt B, a premium increase of Δp imposes a tax burden $B\Delta p$ per period, whose excess burden is minimised by tax smoothing [13]. Distributional consequences are evaluated with social welfare functions with inequality aversion [11]. Pareto efficiency requires that p reflect true marginal risk and not market failure such as asymmetric information [3].

10.19 Warfare economics

In conflict models, war is a state-contingent destruction of capital and a fiscal shock [83, 78]. Let war arrive with hazard $h_w(s)$ and destroy a fraction L_w of the claim. With a mean-reverting hazard, Proposition 8.2 gives the premium

$$p_{\mathrm{war}}(m) = L_w\big[h_\infty + (h_0 - h_\infty)L_1(m;\kappa_p)\big], \tag{44}$$

which is downward sloping when the current hazard h_0 exceeds the long-run level h_∞, as in a crisis, and upward sloping when it is below. War finance, state formation and the fiscal-military state [150, 21, 95, 55] supply the historical evidence; sovereign debt markets have sometimes been used as an indicator of conflict expectations [56].

10.20 Warfare

This section is limited to strategy and economics, with no operational content. In strategic theory, friction and uncertainty are inherent features of conflict [29], and bargaining models explain war as a failure of negotiation under private information or commitment problems [54, 139]. A stylised link to our parameters: higher perceived probability of escalation raises h_w, higher potential damage raises L_w, and prolonged hostilities lengthen the effective horizon over which p_{war} applies. A sovereign's duration of debt (long-dated issuance) reduces rollover risk during crises but locks in the premium priced at issuance.

10.21 Statistics

The estimator is a nonlinear regression with heteroskedastic errors, so [158, 92, 4] apply directly. Model selection for K is by information criteria or out-of-sample loss. A hierarchical (empirical Bayes) extension shares strength across issuers: $\varphi_c \sim \mathcal{N}(\bar{\varphi}, \Sigma)$ for issuer class c, with the posterior mean a precision-weighted compromise between the class estimate and the pooled estimate [79]. Bootstrap inference [47] complements the sandwich estimator for small N.

10.22 Computer science and algorithms

The reduction in Section 6 is an instance of separable nonlinear least squares [66]. Complexity is $O(Nn)$ per function evaluation, so the problem is embarrassingly parallel across bonds [34]. Numerically: use QR rather than normal equations; Gauss–Legendre quadrature for continuous coupon streams; log-sum-exp for long maturities to avoid underflow; automatic differentiation as a check of the analytic Jacobian [123]. Python code appears in Appendix B.

10.23 Data science

The pipeline in Table 6 covers ingestion, cleaning, feature construction, modelling, validation and monitoring. Key risks: look-ahead leakage when computing D_i or filters from future information; survivorship bias from delisted bonds; stale quotes; and concept drift in φ_t. Time-series cross-validation must respect ordering [16].

10.24 Artificial intelligence and machine learning

The class $p(D) = b(D)'\varphi$ can be replaced by a flexible learner: a Gaussian process [130], a gradient-boosted tree, or a neural network [65], trained with stochastic optimisers [99]. Machine-learning methods have been applied to return and bond-risk-premium prediction [68, 17]. Three requirements make such a learner credible here: (i) the sovereign restriction $p_S = 0$ must be imposed as a constraint; (ii) monotonicity or shape constraints should reflect economic theory (e.g. $p(0) = 0$ in the affine case of Proposition 8.1); (iii) uncertainty quantification is needed because the target is an inverse-problem solution. Physics-informed loss terms (e.g. no-arbitrage constraints) improve extrapolation, and regularisation such as ridge or lasso [85, 148] limits overfitting.

10.25 Cybersecurity

Estimation of $\hat{p}$ inherits the integrity risks of its inputs. Benchmark manipulation has occurred [1], and adversarial or poisoned quotes can bias $\hat{\psi}$. Mitigations: (i) bounded-influence estimators (Huber loss) so that one corrupted price has bounded effect [87]; (ii) multiple independent price sources; (iii) cryptographic hashes of input files (e.g. SHA-256 [122]) stored in a model-governance record; (iv) threat modelling [143]; (v) access control on code and data. Cyber incidents also create an issuer-specific component of p, and have measurable market impact [27]; cost-benefit analysis of security investment follows [67, 6].

10.26 Further fields

Actuarial science. Life-contingent and credit-contingent claims share the survival-weighted form of (4); reserves are computed with discount curves and duration matching [20].

Information theory. The change of measure from $\mathbb{P}$ to $\mathbb{Q}$ has relative entropy $\mathbb{E}^{\mathbb{P}}[\ln(\mathrm{d}\mathbb{P}/\mathrm{d}\mathbb{Q})]$, and entropy bounds on the pricing kernel refine (38) [36, 12]. Robust-control formulations interpret part of p as a price of model misspecification [74].

Control and signal processing. Time-varying θ_t, φ_t are filtered by the Kalman filter [94]; the loadings are the observation matrix and the factor dynamics the transition matrix.

Climate economics. Choice of discount rate dominates the valuation of very long-dated damages [146, 124, 157]; $\hat{r_f} + \hat{p}$ at long duration is an observable benchmark for the debate.

Accounting and audit. Fair-value measurement uses observable inputs where possible and model-based inputs otherwise [89]; $\hat{p}$ can be a documented Level 2 or Level 3 input.

History. Interest rates have been recorded for millennia [86], and the sovereign borrowing literature supplies long samples [151, 134].

11 Limitations and research agenda

1. *Scale convention.* The factor $1 + D$ in (1) is a normalisation (Remark 2.1). If Q is a raw price, set $\iota = 0$, or specify the scaling explicitly.
2. *Curve link (A1).* Identifying $r_f(D)$ with the zero yield at maturity D is a modelling choice. Alternatives: rates at the PV-weighted *maturity*, or a key-rate decomposition [84].
3. *Cash-flow risk.* Optionality (calls, puts, sinking funds), inflation indexation and defaults with partial recovery change μ and are not captured by a premium alone.
4. *Nonstationarity.* Structural breaks (regime changes, crises) require time variation.
5. *Weak identification and collinearity* (Remark 5.6).
6. *Normative limits.* $\hat{p}$ is a market price of risk-bearing, not a measure of moral or social value.

Research directions: nonparametric identification with instruments; Bayesian hierarchical models across issuers; no-arbitrage structure imposed on p; sovereign convenience yield as a latent factor; text and geopolitical covariates; adversarially robust estimation.

12 Conclusion

The treatise has derived the continuous-time price and parallel-shift duration of a coupon bond, expressed the implied total discount rate in closed form, and shown that the risk-free and premium components are observationally equivalent in a cross-section unless identified by a common curve and a risk-free sample. A variable-projection nonlinear least-squares estimator with a Nelson–Siegel curve and a basis expansion for $p(D)$ delivers consistent, asymptotically normal estimates with explicit Jacobians and standard errors. Structural models suggest a linear-in-duration or Nelson–Siegel form for p. The multidisciplinary reading shows that duration is a universal first-moment statistic of a decaying process, while the implied premium is a price of whatever the model leaves unexplained, and so must always be read together with its maintained assumptions.

A Additional derivations

A.1 Duration of a continuous coupon with exponential growth

Let $C(s) = C_0 e^{gs'}$, where $s' = s - t$, and constant $r > g$. Then, with $\delta = r - g$,

$$\int_0^\tau C_0 e^{-\delta u} \mathrm{d}u = \frac{C_0}{\delta}(1 - e^{-\delta\tau}), \tag{45}$$

$$\int_0^\tau u C_0 e^{-\delta u} \mathrm{d}u = \frac{C_0}{\delta^2}\big(1 - e^{-\delta\tau}(1 + \delta\tau)\big), \tag{46}$$

and P, D follow from (12)–(13) with $C \mapsto C_0$, $r \mapsto \delta$ in the coupon terms and r in the principal terms. As $\tau \to \infty$ the duration tends to $1/\delta$, so growing coupons lengthen duration.

A.2 Convexity of the constant case

Differentiating (6) twice for constant r, C gives

$$\frac{\partial^2 P_\varepsilon}{\partial \varepsilon^2}\Big|_0 = \int_0^\tau u^2 C e^{-ru}\mathrm{d}u + \tau^2 M e^{-r\tau}, \qquad \int_0^\tau u^2 e^{-ru}\mathrm{d}u = \frac{2 - e^{-r\tau}(2 + 2r\tau + r^2\tau^2)}{r^3}, \tag{47}$$

the second identity by two integrations by parts. Hence $\mathrm{Var}_w(s-t) = P_{\varepsilon\varepsilon}/P - D^2$.

A.3 Why the equation-error regression is inconsistent

Consider the regression $Q_i \sum_k \varphi_k b_k(D_i) = \widetilde{P}_i - Q_i(1 + y_i)$. The regressor $Q_i b_k(D_i)$ contains ε_i through $Q_i = \hat{Q}_i + \varepsilon_i$, and the equation error is $(1+g_i)\varepsilon_i$. Their covariance is $\mathbb{E}[b_k(D_i)(1+g_i)\varepsilon_i^2] \neq 0$. The OLS limit is therefore biased. Dividing the equation by Q_i gives (26), whose regressors are free of ε_i.

A.4 Sensitivity of the premium to the curve

Differentiating $p = \widetilde{P}/Q - 1 - y(D;\theta)$ in θ_0 (a parallel curve shift by h) with $\iota = 1$:

$$\frac{\partial p}{\partial \theta_0} = -\frac{1}{Q}\sum_k m_k(1+m_k)c_k Z_k - 1 - y'(D)\,\partial_{\theta_0} D. \tag{48}$$

The first term dominates for long durations, so an error in the level of the risk-free curve of h maps into an error in $\hat{p}$ of roughly $-\widetilde{P}\,\mathbb{E}_{\tilde{w}}[m]\,h/Q - h$, amplified by the duration-weighted average maturity. This quantifies the joint hypothesis problem for this model.

B Reference implementation (Python)

The listing implements Algorithm 1 (variable projection with E1 and a polynomial basis through the origin). It was run on synthetic data generated from the design in Table 5 and recovered the true parameters to within sampling noise.

```
import numpy as np
from scipy.optimize import least_squares

KAPPA = 2.0

def ns_loadings(m, kappa=KAPPA):
    x = np.maximum(m, 1e-8) / kappa
    L1 = (1.0 - np.exp(-x)) / x
    return np.column_stack([np.ones_like(x), L1, L1 - np.exp(-x)])

def curve_stats(th, times, cfs):
    P, D = [], []
    for m, c in zip(times, cfs):
        z = np.exp(-m * (ns_loadings(m) @ th))
        v = np.sum(c * z)
        P.append(v); D.append(np.sum(m * c * z) / v)
    P, D = np.array(P), np.array(D)
    return P, D, P * (1.0 + D)          # iota = 1

def design(D):
    return np.column_stack([D, D**2])   # premium basis

def reduced_resid(th, Q, isK, times, cfs, sig, return_phi=False):
    P, D, Pt = curve_stats(th, times, cfs)
```

```
    y = ns_loadings(D) @ th             # r_f(D) = y(D; theta)
    g = Pt / Q - 1.0                    # implied sum
    w = Q / (sig * (1.0 + g))           # first-order weights
    B = design(D); wk = w[isK]
    phi = np.linalg.lstsq(B[isK] * wk[:, None],
                          ((g - y)[isK]) * wk, rcond=None)[0]
    p = np.where(isK, B @ phi, 0.0)     # p = 0 on sovereigns
    r = w * (g - y - p)
    return (r, phi) if return_phi else r

def fit(Q, isK, times, cfs, sig, th0=(0.04, -0.01, 0.0)):
    res = least_squares(reduced_resid, np.array(th0), method="lm",
                        args=(Q, isK, times, cfs, sig))
    _, phi = reduced_resid(res.x, Q, isK, times, cfs, sig, True)
    return res.x, phi, res
```

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Glossary

Ambiguity premium Compensation for unknown probabilities, distinct from compensation for known risk.

Arbitrage A riskless profit with zero net investment; its absence restricts prices.

Basis function A fixed function $b_k(D)$ used to expand $p(D)$.

Bootstrap Resampling method for approximating the distribution of an estimator.

Convenience yield The non-pecuniary benefit of holding a safe, liquid asset, which lowers its yield.

Convexity Second-order price sensitivity to rates; here the variance of time under PV weights.

Coupon Periodic (here continuous) interest payment on a bond.

Credit spread Yield of a defaultable bond minus that of a comparable risk-free bond.

Delta method First-order approximation of the variance of a smooth function of an estimator.

Discount factor $Z(t, s)$, the value at t of one unit paid at s.

Dollar duration $P \cdot D$; the currency change in price per unit parallel shift.

Duration (parallel-shift) $-\partial \ln P_\varepsilon / \partial \varepsilon$; the PV-weighted mean time to cash flows.

Essentially affine A term-structure specification in which the market price of risk is affine in the state.

Expected loss $h^{\mathbb{P}} L$; actuarial cost of default.

Fisher–Weil duration Duration with respect to a parallel shift of the entire discount curve.

Forward rate The instantaneous rate implied by the yield curve for a future date.

Generated regressor A regressor computed from estimated parameters.

Hazard rate (intensity) The instantaneous conditional arrival rate of default or another event.

Heteroskedasticity Error variance that varies across observations.

Identification The property that distinct parameters imply distinct distributions of the data.

Implied risk premium A premium obtained by inverting a pricing formula given observed prices.

Indenture The contract governing a bond issue.

Joint hypothesis problem The impossibility of testing a pricing model separately from the assumptions used to apply it.

Levenberg–Marquardt A damped Gauss–Newton algorithm for nonlinear least squares.

Loss given default L; fraction of exposure lost at default.

Market price of risk Compensation per unit of exposure to a risk factor.

Mean residence time Pharmacokinetic analogue of duration.

Nelson–Siegel A three-factor parametric yield-curve family.

Observational equivalence Two parameter values generating the same data distribution.

Pricing kernel (SDF) A positive random variable m with $\mathbb{E}[mR] = 1$ for every traded gross return R.

Present-value weight $w(\mathrm{d}s)$; the share of price attributable to a cash flow.

Risk-neutral measure A probability measure under which discounted prices are martingales.

Sandwich estimator A robust covariance estimator of the form $A^{-1}BA^{-1}$.

Sovereign sample Bonds treated as risk-free, with $p = 0$.

Spread duration Sensitivity of price to a change in the premium, here $1/(1+g)$ in relative terms.

Term premium Compensation for bearing interest-rate risk over a horizon.

Variable projection Elimination of linear parameters from a separable least-squares problem.

Yield curve Yield as a function of maturity.

Zero-coupon bond A bond paying only the terminal amount.

The End